Zhehao Sun

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Education

University of British Columbia (Okanagan Campus) Kelowna, British Columbia, Canada
Master of Arts in Interdisciplinary Graduate Studies (Digital Arts and Humanities)
GPA: 91.0% (A+) May 2025 – May 2027 (expected)
Supervisor: Dr. Megan Smith *Committee:* Dr. Miles Thorogood, Dr. Patricia Lasserre

Savannah College of Art and Design (SCAD) Savannah, Georgia, USA
Bachelor of Fine Arts in Interactive Design and Game Development September 2015 – June 2018
Transferred from University of North Georgia (*Studio Art Major*, January 2013 – May 2015)

Research Interests

Game mechanics and player experience (PX); procedural content generation (PCG) and AI-assisted game design

Publications

Journal Articles

1. Xu, Y., **Sun, Z.**, Zhen, C., Lin, Y.-S., Sarker, T. H., Thorogood, M., Lasserre, P., & Dulic, A. (2026). From engagement to resilience: A systematic review of game-based learning for environmental resilience. *Sustainability*, 18(5), 2305. https://doi.org/10.3390/su18052305

Refereed Conference Proceedings & Book Chapters

1. **Sun, Z.**, Xu, Y., Zhen, C., & Smith, M. (2026). "You Were Never Meant to Win": A taxonomy of scripted loss in digital games. In *Companion Proceedings of the Annual Symposium on Computer-Human Interaction in Play (ACM CHI PLAY 2026)*. [Work-in-Progress] https://doi.org/10.1145/3800965.3834328

2. **Sun, Z.** (Presenter), Xu, Y., Lasserre, P., Dulic, A., & Smith, M. (2026). Re-evaluating hyper-realism in virtual reality game design through the embodied simulation gap. In *Proceedings of the 2026 Digital Games Research Association Conference (DiGRA 2026)*. Ireland. [Extended abstract] https://doi.org/10.26503/dl.v2026i4.2997

3. **Sun, Z.** (Presenter), Xu, Y., Zhen, C., Lin, Y.-S., Thorogood, M., Lasserre, P., Dulic, A., & Smith, M. (2026). Struggle as flow: Challenge, design, and experience in Soulslike games. In *Lecture Notes in Computer Science (HCI International 2026)*. Springer, Cham. https://doi.org/10.1007/978-3-032-30405-6_15

4. Zhen, C., Xu, Y., **Sun, Z.**, Dulic, A., & Thorogood, M. (2026). Sound as the score: Designing progressive audio layering as embodied music reward. In *Companion Proceedings of the Annual Symposium on Computer-Human Interaction in Play (ACM CHI PLAY 2026)*. [Work-in-Progress] https://doi.org/10.1145/3800965.3834330

5. Xu, Y., **Sun, Z.**, Lin, Y.-S., Lasserre, P., & Dulic, A. (2026). The absent teacher in eco-games: A theoretical framework for teacher integration in eco-games. In *Proceedings of the 2026 Digital Games Research Association Conference (DiGRA 2026)*. Ireland. [Extended abstract] https://doi.org/10.26503/dl.v2026i4.3002

6. Xu, Y., **Sun, Z.**, Zhen, C., Lin, Y.-S., Thorogood, M., Smith, M., Lasserre, P., & Dulic, A. (2026). Mapping ecological empathy: A semantic network analysis of player perceptions in 3D environmental education games. In *Lecture Notes in Computer Science (HCI International 2026)*. Springer, Cham. https://doi.org/10.1007/978-3-032-30405-6_16

Teaching & Academic Experience

University of British Columbia (Okanagan Campus) Kelowna, British Columbia, Canada
Teaching Assistant, CCS 325: AI & Creativity September 2026 – December 2026
About 200 students, 2 TAs. Lead tutorial sessions, grade, give student feedback, and facilitate in-class activities on generative AI, prompt engineering and creative coding.

Marker, CCS 330: Introduction to Game Design December 2025 – February 2026
About 120 students. Graded assignments and gave written feedback.

Teaching Assistant, ENGL 155 / DIHU 155 August 2025 – April 2026
Writing and Making with Technology in the Humanities; about 140 students, 3 TAs. Independently led a weekly mandatory discussion session, gave formative (ungraded) feedback on essays, and built the tutorial materials website.

Selected Projects

Wastescape – Research-Creation Roguelite Game (MA Thesis Project) May 2025 – Present
University of British Columbia (Okanagan Campus), Canada | Playable build on itch.io

Designed and built a playable 2D excavation roguelite game in Unity 6.5 (C#) and an early HTML5/JavaScript prototype.

Implemented deterministic procedural terrain generation, grid locomotion physics, hazard chain reactions, and an artifact archive system.

Beaver World – 3D Educational Game (Work Study) May 2026 – Present
Technical Artist / Developer | University of British Columbia (Okanagan Campus), Canada

Built a procedural vegetation generator (Unity editor tool) that fills five 4 km^2 HDRP scenes with trees, shrubs and grass from per-scene ecological rules, replacing manual placement across the game.

Fixed a severe frame-rate drop caused by over-dense terrain detail: profiled GPU passes and wrote a mesh reducer that cut grass meshes by about 87% while keeping wind-animation vertex data, reducing triangles per frame from 255 M to about 40 M.

Removed a multi-second scene-load stall and halved texture memory (3.5 GB to 1.7 GB) by baking reflection probes and removing per-object scene lookups.

Created the wildfire visuals and post-fire environment (fire spread, burnt-tree and burnt-ground materials, per-cycle regrowth); built the main menu and the shared settings system.

Destination: Imagination – VR Entertainment Ride Project January 2018 – June 2018
Digital Artist | Savannah College of Art and Design, USA

Built a VR amusement ride experience for the 2018 SCAD Day public exhibition (Featured by Unreal Engine).

Designed interaction flow, navigation, and gameplay logic in Unreal Engine Blueprints; produced optimized 3D assets and VFX for real-time VR performance.

Professional Experience

Art First International Education (New Channel International Education Group) China
Digital Media Instructor October 2019 – April 2025

Taught Unreal Engine development, computer graphics fundamentals, and game design principles; guided portfolio and demo reel development for students applying to international universities.

CATS Co., Ltd. (NTT Data) Shinyokohama, Japan
3D Designer September 2018 – September 2019

Contributed to an autonomous vehicle technology visualization project through cross-functional collaboration with programming and technical teams.

Developed real-time 3D data visualization systems using Unreal Engine for automotive sensor data representation.

Implemented web-based 3D rendering solutions utilizing Java, WebGL, and C# technologies.

Xi'an Kushi Information Technology Co., Ltd. Xi'an, China
3D Artist Intern June 2018 – September 2018

Participated in augmented reality application development using Unity and Vuforia platforms.

Created 3D models and textures for virtual environment assets.

Technical Skills

Game Engines: Unreal Engine (Blueprints, custom shaders, real-time VFX), Unity (C#, UI Toolkit, 2D/3D/VR), Vuforia
Programming & Web: C#, JavaScript / HTML5, Three.js / WebGL, Java
3D Production: Modeling, sculpting, retopology, UV mapping, texturing, rigging, animation, environment lighting
Research & Design: Game design, player experience, computer graphics, UX for interactive media

Honors & Awards

UBC Okanagan Graduate Research Scholarship (OGRS) ($5,000/year; $10,000 total)	2025 – 2027
SCAD Dean's List	2017 – 2018
SCAD Academic Scholarship ($10,000/year)	2015 – 2018
SCAD Student Incentive Scholarship	2015 – 2018